

1. **What is the voltage drop at C?**

Answer:

VDF for 4/0, 3-ph = 0.92

VDF for 1/0, 3-ph = 1.55

$$VD_C = \frac{200 \times 5 \times 1.55}{1000} + \frac{500 \times 4 \times 0.92}{1000}$$

$$= (1.55+1.84)\ V$$

$$= 3.39\ \text{Volt.}$$

Voltage Level and Calculation

For design it is consider that Maximum Voltage Drop will be 3 % for any conductor and P.F is 0.95

For 4/0 ACSR, 3-Ph HT :

$VD = (KW \times S \times VDF) \div 1000$

Or, $3\%\ 6350 = (KW \times 1 \times 0.92) \div 1000$

(At 230V base)

Or, $\{(0.03 \times 6350 \times 230) \div 6350\} = \{(KW \times 1 \times 0.92) \div 1000\}$

Or, $KW\text{-}KM = (0.03 \times 6350 \times 1000 \times 230) \div (6350 \times 0.92)$

$= 7500$

So, $KVA\text{-}KM = 7500 \div 0.95$

$= 7894$

Hence , $MVA\text{-}KM = 7.894 \approx 8$

Voltage Level and Calculation

For #3 ACSR, 1-Ph LT :

$VD = IZ$

Or, $3\ \% \ 230 = (W \div V\cos\theta) \times (R\cos\theta + X\sin\theta)$

Or, $0.03 \times 230 = \{W \div (230 \times 0.95)\} \times \{(1.3946 \times 0.95) + (.765 \times .312)\}$

Or, $6.9 = (W \times 1.56) \div (230 \times 0.95)$

Or, $W\text{-}KM = (230 \times 0.95 \times 6.9) \div 1.56$

$= 966.44$

Or, $KW\text{-}KM = 0.966$

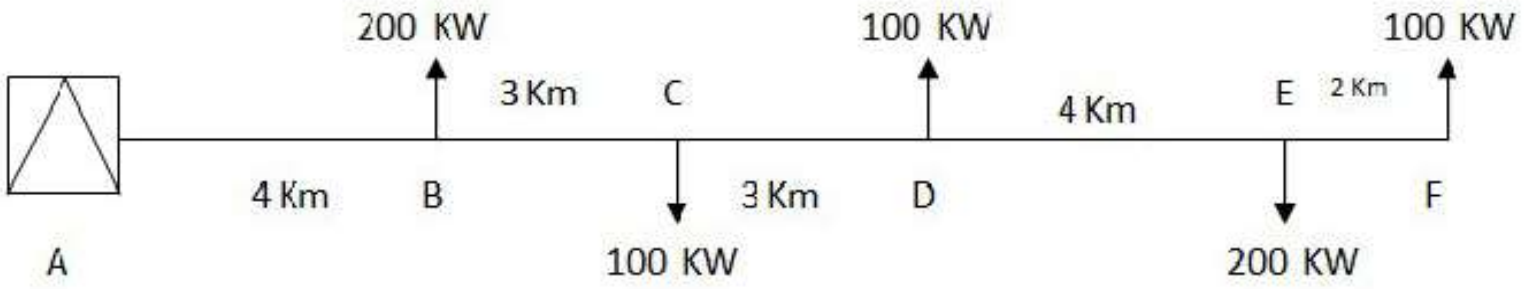
Or, $KVA\text{-}KM = (0.966 \div 0.95)$

$= 1.01 \approx 1$

Voltage Level and Calculation

MVA/KVA-KM Table

Sl No.	Size of Conductor	MVA/KVA-Km
1	4/0 ACSR – 3-ϕ HT	8 MVA - Km
2	1/0 ACSR – 3-ϕ HT	5 MVA - Km
3	# 3 ACSR – 3-ϕ HT	2.5 MVA - Km
4	# 3 ACSR – 1-ϕ HT	800 KVA - Km
5	# 3 ACSR – 3-ϕ LT	2.5 KVA - Km
6	# 3 ACSR – 1-ϕ LT	1 KVA - Km
7	# 3 Quadruplex – LT	2.5 KVA - Km
8	# 3 Duplex – LT	1 KVA - Km
9	1/0 Quadruplex – LT	5 KVA – Km
10	477 MCM ACSR – 3-ϕ HT	16 MVA - Km



KW-KM = (100 x 2) + (300 x 4) + (400 x 3) + (500 x 3) + (700 x 4)
= 200 + 1200 + 1200 + 1500 + 2800
= 6900

KVA-KM= 6900/0.95 = 7263

MVA-KM = 7.26

Therefore, **Conductor Size = 4/0 ACSR**

Voltage Level and Calculation

Questions????